Machine Learning Fundamentals

Source: Open technical documentation and standard references (public knowledge, paraphrased)

This document gives a short overview of Machine Learning Fundamentals.

Supervised learning trains a model on labelled examples, while unsupervised learning finds patterns in unlabelled data. Overfitting happens when a model learns the training data too closely and performs poorly on new data. Underfitting happens when a model is too simple to capture the pattern in the data.

A common train-test split keeps 80 percent of the data for training and 20 percent for testing. Precision is the share of predicted positives that are correct, and recall is the share of actual positives that are found. K-fold cross-validation splits the data into k parts and trains and tests the model k times.

Regularisation adds a penalty on model complexity to reduce overfitting. Gradient descent updates model parameters step by step in the direction that reduces the loss.

Note: This is a simplified educational summary written in our own words from public information. It is not legal, financial or medical advice. Verify details with the official source before relying on them.